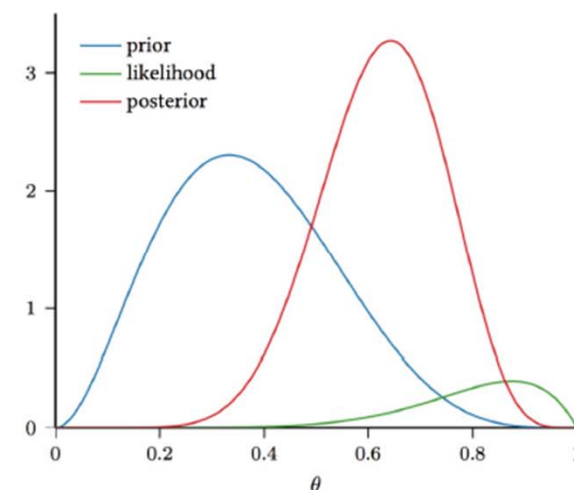


Table of Contents

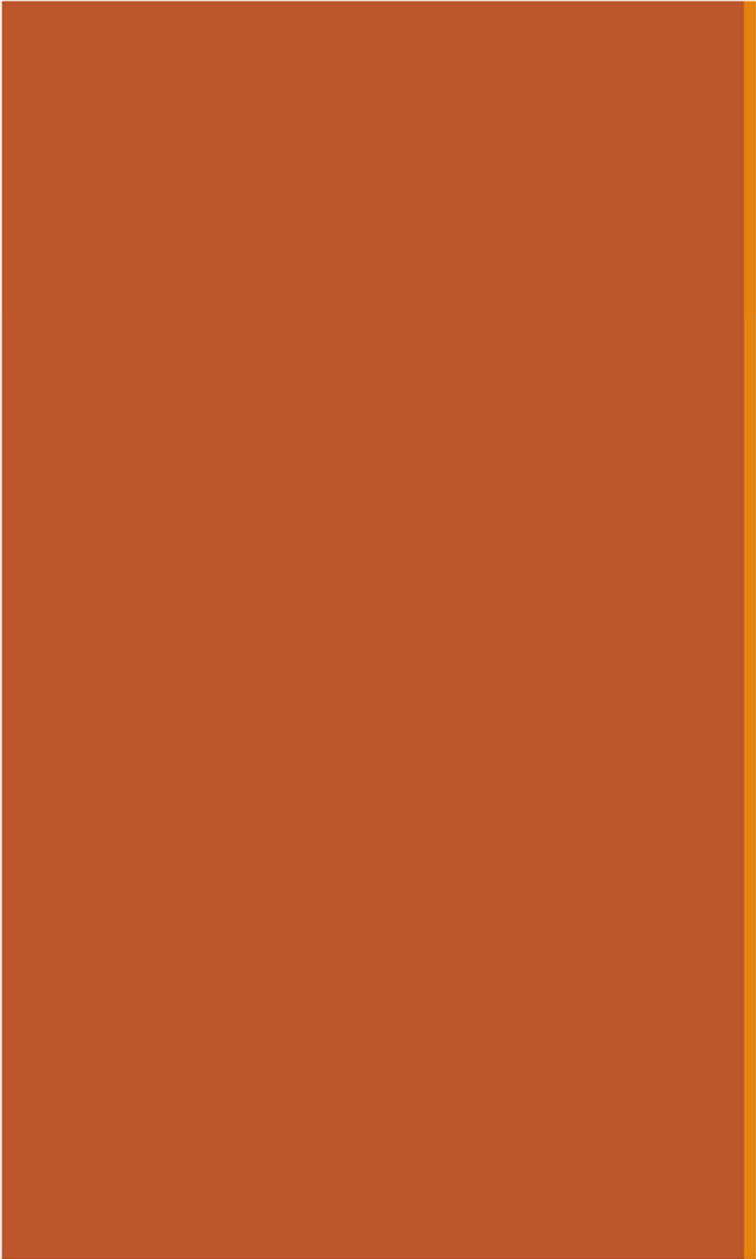
2	Maximum a Posteriori
10	Bernoulli MAP: Clumsy Prior
14	Bernoulli MAP: Conjugate Prior
23	Choosing Hyperparameters
29	Extra: Other Conjugates



22: MAP

Jerry Cain
March 1, 2024

[Lecture Discussion on Ed](#)

A solid orange horizontal bar spanning the width of the slide, positioned at the top.

Maximum a Posteriori Estimator

Maximum Likelihood Estimator

Review

Consider a sample of n iid random variables $X_1, X_2, \dots, X_n$.

Maximum Likelihood Estimator (MLE) What parameter θ **maximizes the likelihood** of our observed data $(X_1, X_2, \dots, X_n)$?

$$L(\theta) = f(X_1, X_2, \dots, X_n | \theta) \\ = \prod_{i=1}^n f(X_i | \theta)$$

$$\theta_{MLE} = \arg \max_{\theta} f(X_1, X_2, \dots, X_n | \theta)$$

likelihood of data

Observations:

- MLE determines θ value that maximizes the probability of observing the sample.
- If we're estimating θ , couldn't we just **maximize the probability of θ** ?

Today: **Bayesian estimation** using the Bayesian definition of probability!

Maximum A Posteriori (MAP) Estimator

Not Review! New!

Consider a sample of n iid random variables $X_1, X_2, \dots, X_n$.

Maximum
Likelihood
Estimator
(MLE)

What parameter θ
maximizes the likelihood
of our observed data
($X_1, X_2, \dots, X_n$)?

$$L(\theta) = f(X_1, X_2, \dots, X_n | \theta) \\ = \prod_{i=1}^n f(X_i | \theta)$$

$$\theta_{MLE} = \arg \max_{\theta} f(X_1, X_2, \dots, X_n | \theta)$$

likelihood of data

Maximum
a Posteriori
(MAP)
Estimator

Given the sample data
($X_1, X_2, \dots, X_n$),
what is the **most probable**
parameter θ ?

$$\theta_{MAP} = \arg \max_{\theta} f(\theta | X_1, X_2, \dots, X_n)$$

posterior distribution
of θ

Maximum A Posteriori (MAP) Estimator

Consider a sample of n iid random variables $X_1, X_2, \dots, X_n$.

def The **Maximum a Posteriori (MAP) Estimator** of θ is the value of θ that maximizes the posterior distribution of θ .

$$\theta_{MAP} = \arg \max_{\theta} f(\theta | X_1, X_2, \dots, X_n)$$

Intuition with Bayes' Theorem:

$L(\theta)$, probability of data
given parameter θ

notice that
both the prior
and the posterior
are focusing on
 θ !

After seeing
data, posterior
belief of θ

posterior

$P(\theta | \text{data})$

likelihood prior

$$= \frac{P(\text{data} | \theta) P(\theta)}{P(\text{data})}$$

Before seeing data,
prior belief of θ

Solving for θ_{MAP}

- Observe data: $X_1, X_2, \dots, X_n$, all iid
- Let likelihood be same as MLE: $f(X_1, X_2, \dots, X_n | \theta) = \prod_{i=1}^n f(X_i | \theta)$
- Let the prior distribution of θ be $g(\theta)$.

$$\begin{aligned}
 \theta_{MAP} &= \arg \max_{\theta} \overset{\text{posterior distribution}}{f(\theta | X_1, X_2, \dots, X_n)} = \arg \max_{\theta} \frac{\overset{\text{likelihood function}}{f(X_1, X_2, \dots, X_n | \theta)} \overset{\text{prior distribution}}{g(\theta)}}{\underset{\text{data distribution}}{h(X_1, X_2, \dots, X_n)}} \quad (\text{Bayes' Theorem}) \\
 &= \arg \max_{\theta} \frac{g(\theta) \prod_{i=1}^n f(X_i | \theta)}{h(X_1, X_2, \dots, X_n)} \quad \theta \text{와 관련없음 (independence)} \\
 &= \arg \max_{\theta} g(\theta) \prod_{i=1}^n f(X_i | \theta) \quad (1/h(X_1, X_2, \dots, X_n) \text{ is a positive constant w.r.t. } \theta) \\
 &= \arg \max_{\theta} \left(\log g(\theta) + \sum_{i=1}^n \log f(X_i | \theta) \right) \quad \text{log 변환}
 \end{aligned}$$



θ_{MAP} : Interpretation 1

- Observe data: $X_1, X_2, \dots, X_n$, all iid
- Let likelihood be same as MLE: $f(X_1, X_2, \dots, X_n | \theta) = \prod_{i=1}^n f(X_i | \theta)$
- Let the prior distribution of θ be $g(\theta)$.

$$\theta_{MAP} = \arg \max_{\theta} f(\theta | X_1, X_2, \dots, X_n) = \arg \max_{\theta} \frac{f(X_1, X_2, \dots, X_n | \theta) g(\theta)}{h(X_1, X_2, \dots, X_n)} \quad (\text{Bayes' Theorem})$$

$$= \arg \max_{\theta} \frac{g(\theta) \prod_{i=1}^n f(X_i | \theta)}{h(X_1, X_2, \dots, X_n)} \quad (\text{independence})$$

$$= \arg \max_{\theta} g(\theta) \prod_{i=1}^n f(X_i | \theta) \quad (1/h(X_1, X_2, \dots, X_n) \text{ is a positive constant w.r.t. } \theta)$$

$$= \arg \max_{\theta} \left(\log g(\theta) + \sum_{i=1}^n \log f(X_i | \theta) \right)$$

θ_{MAP} maximizes
log prior + log-likelihood

θ_{MAP} : Interpretation 2

- Observe data: $X_1, X_2, \dots, X_n$, all iid
- Let likelihood be same as MLE: $f(X_1, X_2, \dots, X_n | \theta) = \prod_{i=1}^n f(X_i | \theta)$
- Let the prior distribution of θ be $g(\theta)$.

$$\begin{aligned}\theta_{MAP} &= \arg \max_{\theta} f(\theta | X_1, X_2, \dots, X_n) = \arg \max_{\theta} \frac{g(\theta) \prod_{i=1}^n f(X_i | \theta)}{h(X_1, X_2, \dots, X_n)} && \text{The mode of the posterior distribution of } \theta \text{ (Bayes' Theorem)} \\ &= \arg \max_{\theta} \frac{g(\theta) \prod_{i=1}^n f(X_i | \theta)}{h(X_1, X_2, \dots, X_n)} && \text{(independence)} \\ &= \arg \max_{\theta} g(\theta) \prod_{i=1}^n f(X_i | \theta) && (1/h(X_1, X_2, \dots, X_n) \text{ is a positive constant w.r.t. } \theta) \\ &= \arg \max_{\theta} \left(\log g(\theta) + \sum_{i=1}^n \log f(X_i | \theta) \right) && \theta_{MAP} \text{ maximizes log prior + log-likelihood}\end{aligned}$$

Mode: A statistic of a random variable

The **mode** of a random variable X is defined as:

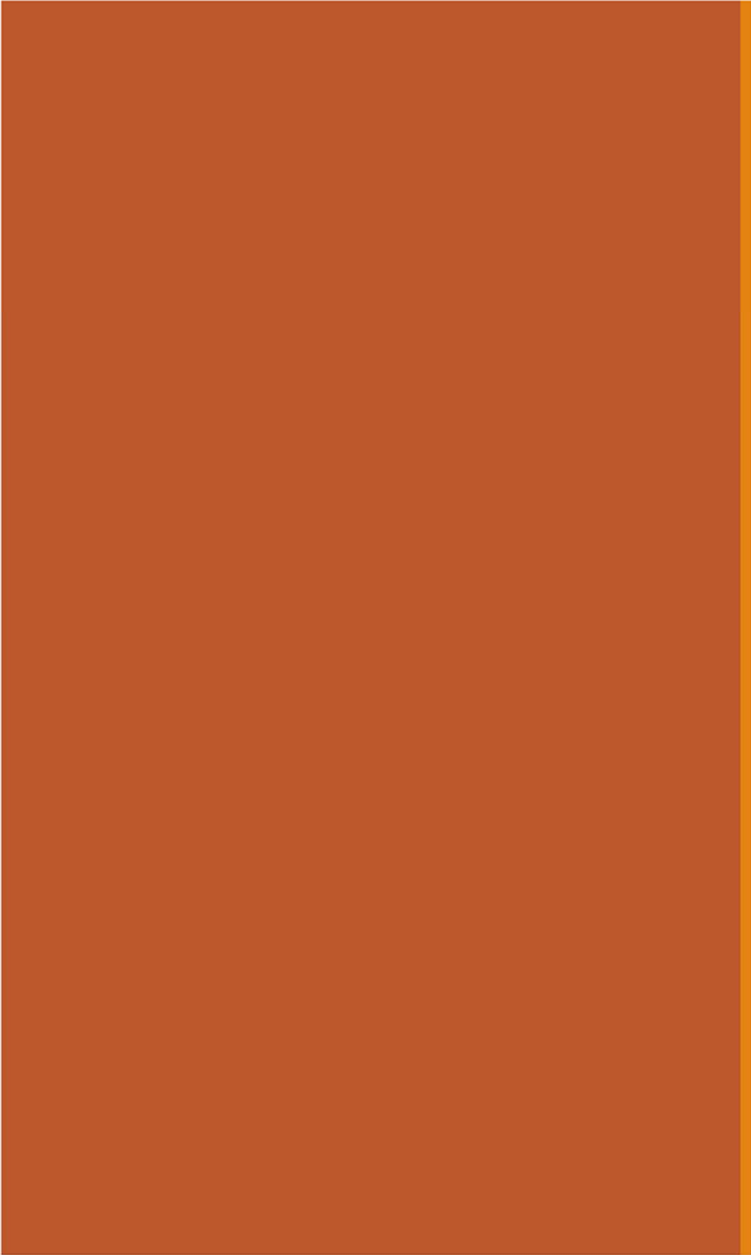
$$\begin{array}{ll} (X \text{ discrete, PMF } p(x)) & \arg \max_x p(x) \\ & \arg \max_x f(x) \quad (X \text{ continuous, PDF } f(x)) \end{array}$$

- Intuitively: The value of X that is "most likely".
- Note that some distributions may not have a unique mode (e.g., Uniform distribution, or Bernoulli(0.5))

$$\theta_{MAP} = \arg \max_{\theta} f(\theta | X_1, X_2, \dots, X_n)$$



θ_{MAP} is the most likely θ given the data $X_1, X_2, \dots, X_n$.

A solid orange horizontal bar spanning the width of the slide, positioned at the top.

Bernoulli MAP: Choosing a prior

How does MAP work? (for Bernoulli)

Observe data

Choose model

Choose **prior** on θ

Find $\theta_{MAP} =$
 $\arg \max_{\theta} f(\theta | X_1, X_2, \dots, X_n)$

n heads, m tails

Bernoulli(p)

(some $g(\theta)$)

maximize
log prior + **log-likelihood**

$$\log g(\theta) + \sum_{i=1}^n \log f(X_i | \theta)$$

- Differentiate, set to 0
- Solve

MAP depends on what $g(\theta)$ we choose.

MAP for Bernoulli

- Flip a coin 8 times. Observe $n = 7$ heads and $m = 1$ tail.
- Choose a prior on θ . What is θ_{MAP} ?

Suppose we pick a prior $\theta \sim \mathcal{N}(0.5, 1^2)$. $g(\theta) = \frac{1}{\sqrt{2\pi}} e^{-(p-0.5)^2/2}$

this seems reasonable — it's centered around 0.5 but allows for a belief that there's bias in one direction.

1. Determine log prior + log likelihood

$$\log g(\theta) + \log f(X_1, X_2, \dots, X_n | \theta)$$

$$= \log \left(\frac{1}{\sqrt{2\pi}} e^{-(p-0.5)^2/2} \right) + \log \left(\binom{n+m}{n} p^n (1-p)^m \right)$$

$$= -\log(\sqrt{2\pi}) - (p-0.5)^2/2 + \log \binom{n+m}{n} + n \log p + m \log(1-p)$$

2. Differentiate wrt (each) θ , set to 0

$$-(p-0.5) + \frac{n}{p} - \frac{m}{1-p} = 0$$

We should choose a prior that's easier to deal with. This one is hard!

3. Solve resulting equations

cubic equations, nope not going to do it

reasonable counterreaction: are we justified in avoiding this $g(\theta)$ because it's difficult.

A better approach: Use conjugate distributions

Observe data

Choose model

Choose **prior on θ**

Find $\theta_{MAP} =$
 $\arg \max_{\theta} f(\theta | X_1, X_2, \dots, X_n)$

n heads, m tails

Bernoulli(p)

if it's up to us, choose something that's sensible and easy to manipulate.

(some $g(\theta)$)

(choose conjugate distribution)

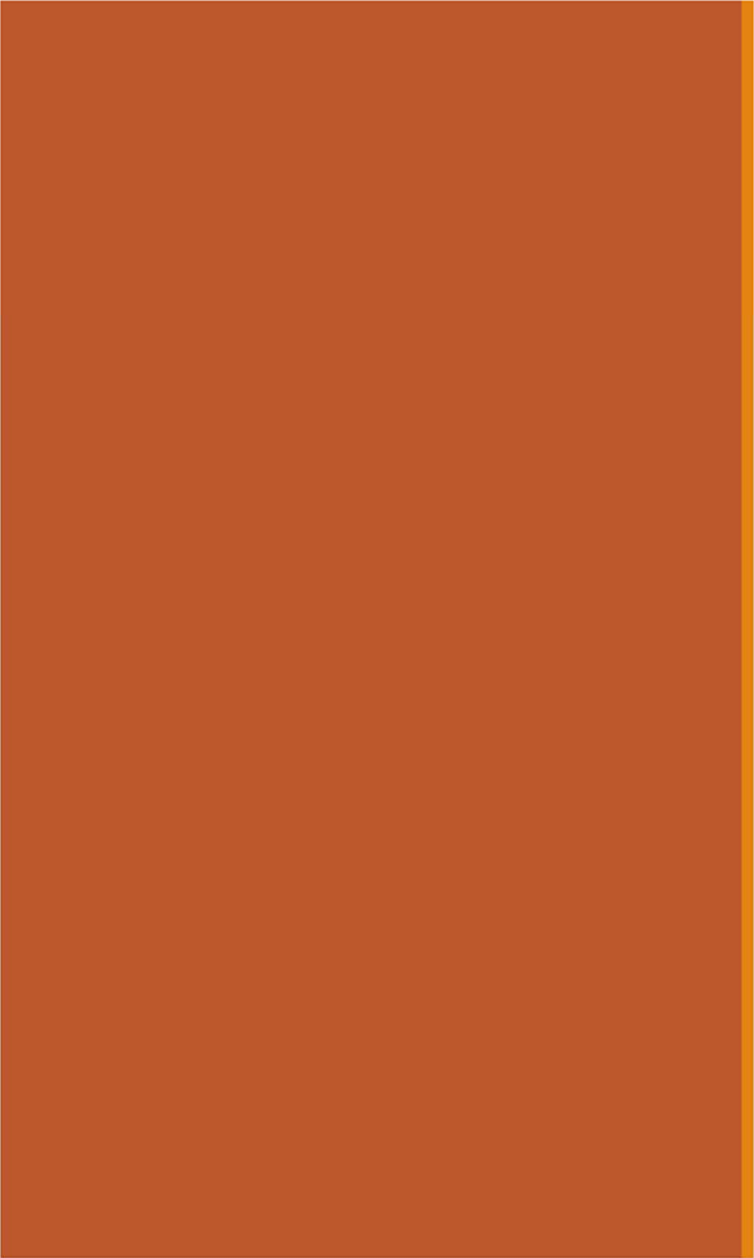
maximize
log prior + log-likelihood

$$\log g(\theta) + \sum_{i=1}^n \log f(X_i | \theta)$$

- Differentiate, set to 0
- Solve



Up next: Conjugate priors are great for MAP!

A solid orange horizontal bar spanning the width of the slide, positioned at the top.

Bernoulli MAP: Conjugate prior

Beta is a conjugate distribution for Bernoulli

Mostly Review

Beta is a **conjugate distribution** for Bernoulli, meaning:

- Prior and posterior parametric forms are the same
- Practically, conjugate means easy update:
Add numbers of "successes" and "failures" seen to Beta parameters.
- You can set the prior to reflect how fair/biased you think the experiment is a priori.

Prior Beta($a = n_{imag} + 1, b = m_{imag} + 1$)

Experiment Observe n successes and m failures

Posterior Beta($a = n_{imag} + n + 1, b = m_{imag} + m + 1$)

Mode of Beta(a, b): $\frac{a - 1}{a + b - 2}$

(we'll prove this in a few minutes)

Beta parameters a, b are called **hyperparameters**.
Interpret Beta(a, b): $a + b - 2$ trials,
of which $a - 1$ are successes

How does MAP work? (for Bernoulli)

Observe data

Choose model

Choose **prior on θ**

Find $\theta_{MAP} =$
 $\arg \max_{\theta} f(\theta | X_1, X_2, \dots, X_n)$

n heads, m tails

Bernoulli(p)



(some $g(\theta)$)

(choose conjugate distribution)

maximize
log prior + log-likelihood

Mode of posterior
distribution of θ

$$\log g(\theta) + \sum_{i=1}^n \log f(X_i | \theta)$$

(posterior is also conjugate)

- Differentiate, set to 0
- Solve

Conjugate strategy: MAP for Bernoulli

- Flip a coin 8 times. Observe $n = 7$ heads and $m = 1$ tail. } Define as data, D
- Choose a prior on θ . What is θ_{MAP} ?

1. Choose a prior

Suppose we pick a prior $\theta \sim \text{Beta}(a, b)$.

2. Determine posterior

Because Beta is a conjugate distribution for Bernoulli, the posterior distribution is $\theta | D \sim \text{Beta}(a + n, b + m)$

3. Compute MAP

$$\theta_{MAP} = \frac{a + n - 1}{a + n + b + m - 2} \quad (\text{mode of } \text{Beta}(a + n, b + m))$$



MAP in practice

- Flip a coin 8 times. Observe $n = 7$ heads and $m = 1$ tail.
- What is the MAP estimator of the Bernoulli parameter p , if we assume a prior on p of $\text{Beta}(2, 2)$?

1. Choose a prior

$$\theta \sim \text{Beta}(2, 2).$$

mode of prior $\text{Beta}(2, 2)$

$$\frac{2-1}{2+2-2} = \frac{1}{2}$$



Before flipping the coin, we imagined 2 trials: 1 imaginary head, 1 imaginary tail.

9 is really a 2+7
3 is really a 2+1

2. Determine posterior

Posterior distribution of θ given observed data is $\text{Beta}(9, 3)$

3. Compute MAP

$$\theta_{\text{MAP}} = \frac{8}{10}$$

mode = $\frac{9-1}{9+3-2} = \frac{8}{10}$

After the experiment, we saw 10 trials: 8 heads (imaginary and real), 2 tails (imaginary and real).

Proving the mode of Beta

Observe data

Choose model

Choose prior on θ

Find $\theta_{MAP} = \arg \max_{\theta} f(\theta|X_1, X_2, \dots, X_n)$

These are **equivalent interpretations** of θ_{MAP} .
We'll use this equivalence to prove the mode of Beta.

n heads, m tails

Bernoulli(p)

(some arbitrary $g(\theta)$)

(choose conjugate)
Beta(a, b)

maximize
log prior + log-likelihood

$$\log g(\theta) + \sum_{i=1}^n \log f(X_i|\theta)$$

- Differentiate, set to 0
- Solve

**Mode of posterior
distribution of θ**

(posterior is also
conjugate)

From first principles: MAP for Bernoulli, conjugate prior

- Flip a coin 8 times. Observe $n = 7$ heads and $m = 1$ tail.
- Choose a prior on θ . What is θ_{MAP} ?

Suppose we pick a prior $\theta \sim \text{Beta}(a, b)$. $g(\theta = p) = \frac{1}{\beta} p^{a-1} (1-p)^{b-1}$ normalizing constant, β

1. Determine log prior + log likelihood

$$\begin{aligned} \log g(\theta) + \log f(X_1, X_2, \dots, X_n | \theta) &= \log \left(\frac{1}{\beta} p^{a-1} (1-p)^{b-1} \right) + \log \left(\binom{n+m}{n} p^n (1-p)^m \right) \\ &= \log \frac{1}{\beta} + \underbrace{(a-1) \log(p)}_{\text{green}} + \underbrace{(b-1) \log(1-p)}_{\text{cyan}} + \log \binom{n+m}{n} + \underbrace{n \log p}_{\text{red}} + \underbrace{m \log(1-p)}_{\text{magenta}} \end{aligned}$$

2. Differentiate w.r.t. (each) θ , set to 0

$$\underbrace{\frac{a-1}{p}}_{\text{green}} + \underbrace{\frac{n}{p}}_{\text{red}} - \underbrace{\frac{b-1}{1-p}}_{\text{cyan}} - \underbrace{\frac{m}{1-p}}_{\text{magenta}} = 0$$

3. Solve (next slide)

From first principles: MAP for Bernoulli, conjugate prior

- Flip a coin 8 times. Observe $n = 7$ heads and $m = 1$ tail.
- Choose a prior θ . What is θ_{MAP} ?

Suppose we pick a prior $\theta \sim \text{Beta}(a, b)$. $g(\theta) = \frac{1}{\beta} p^{a-1} (1-p)^{b-1}$ normalizing
constant, β

3. Solve for p $\frac{a-1}{p} + \frac{n}{p} - \frac{b-1}{1-p} - \frac{m}{1-p} = 0$ (from previous slide)

$$\Rightarrow \frac{a+n-1}{p} - \frac{b+m-1}{1-p} = 0$$

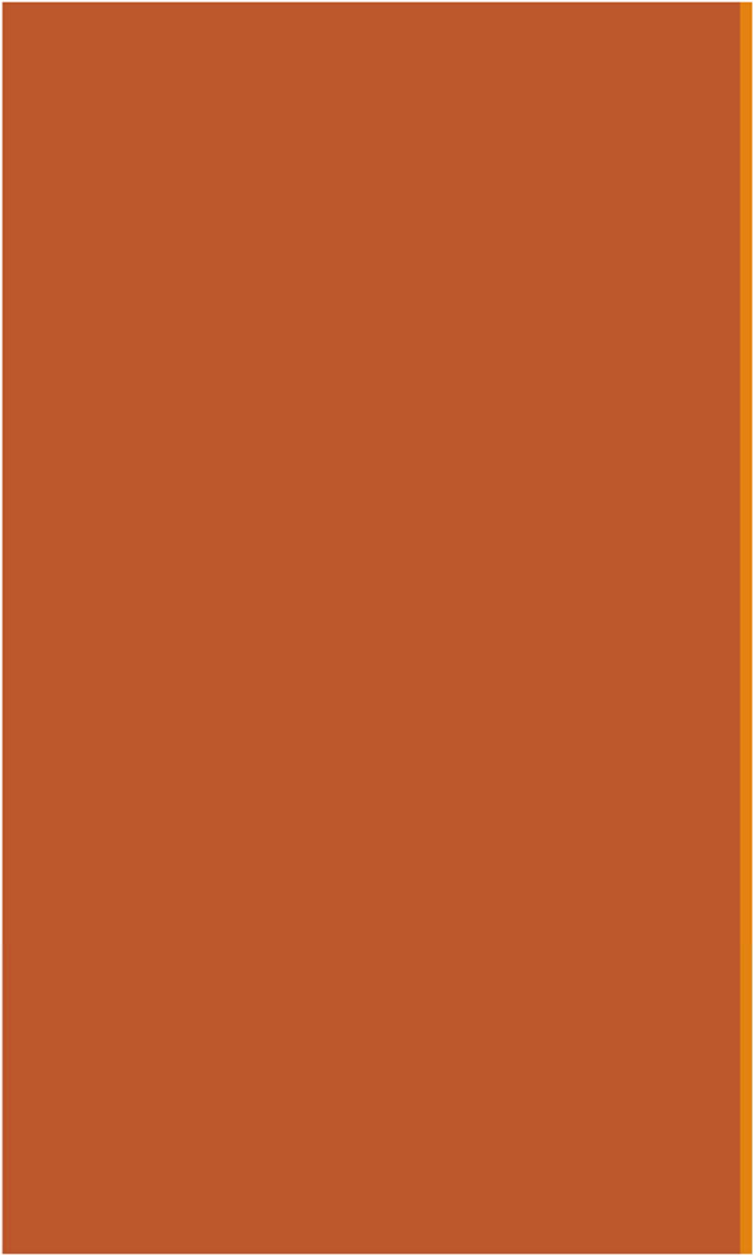
$$\Rightarrow (a+n-1) - (a+n-1)p = (b+m-1)p$$

$$\Rightarrow p(a+n+b+m-2) = a+n-1$$

$$\theta_{MAP} = \frac{a+n-1}{a+n+b+m-2} \quad \checkmark$$

The mode of the posterior,
 $\text{Beta}(a+n, b+m)$!

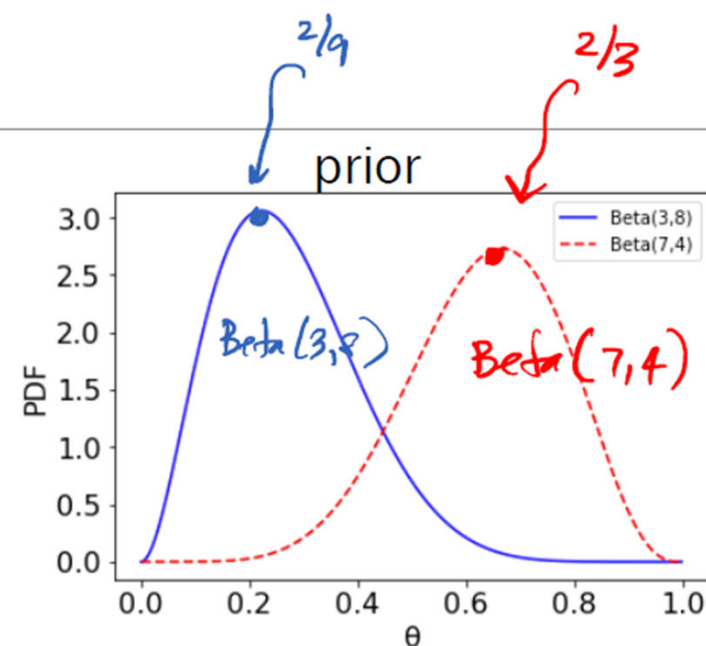
If we choose a conjugate prior, we avoid calculus with MAP, and we can simply report mode of posterior.

A solid orange horizontal bar spanning the width of the slide, with a thin yellow vertical line on its right side.

Choosing hyperparameters for conjugate prior

Where'd you get them priors?

- Let θ be the probability a coin turns up heads.
- Model θ with 2 different priors:
 - Prior 1: $\text{Beta}(3,8)$: 2 imaginary heads, 7 imaginary tails mode: $\frac{2}{9}$
 - Prior 2: $\text{Beta}(7,4)$: 6 imaginary heads, 3 imaginary tails mode: $\frac{6}{9}$



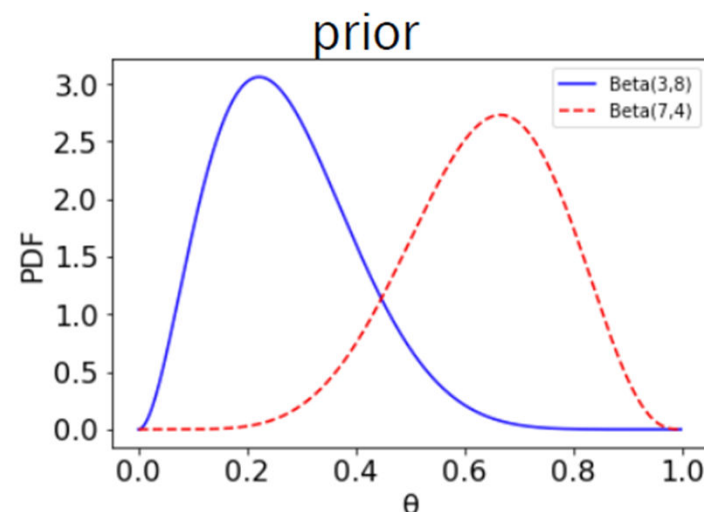
Now flip 100 coins and get 58 heads and 42 tails.

1. What are the two posterior distributions?
2. What are the modes of the two posterior distributions?



Where'd you get them priors?

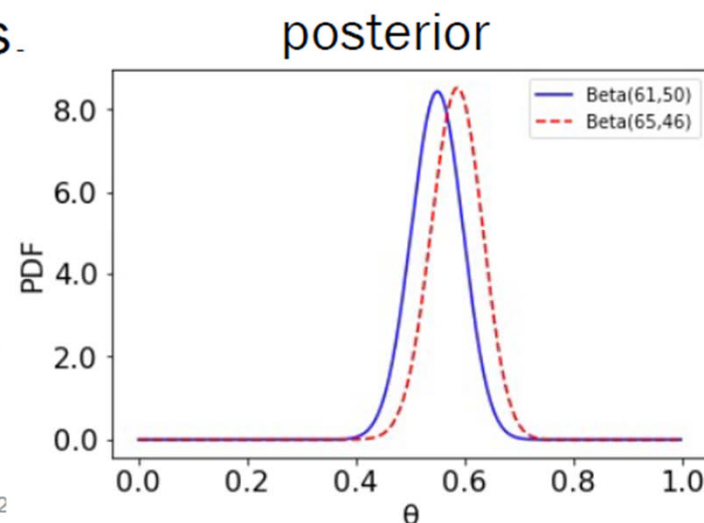
- Let θ be the probability a coin turns up heads.
- Model θ with 2 different priors:
 - Prior 1: **Beta(3,8)**: 2 imaginary heads, 7 imaginary tails mode: $\frac{2}{9}$
 - Prior 2: **Beta(7,4)**: 6 imaginary heads, 3 imaginary tails mode: $\frac{6}{9}$



Now flip 100 coins and get 58 heads and 42 tails.

Posterior 1: **Beta(61,50)** mode: $\frac{60}{109}$

Posterior 2: **Beta(65,46)** mode: $\frac{64}{109}$



Provided we collect enough data, posteriors will converge to the true value and choice of priors will matter less and less.

Laplace smoothing

MAP with **Laplace smoothing**: a prior which represents k imagined observations of each outcome.

- Categorical data (i.e., Multinomial, Bernoulli/Binomial)
- Also known as additive smoothing

Laplace estimate Imagine $k = 1$ of each outcome
(follows from Laplace's "[law of succession](#)")

Example: Laplace estimate for probabilities from previously mentioned experiment (100 coins: 58 heads, 42 tails)

$$\text{heads } \frac{59}{102} = \frac{58+1}{100+2} \quad \text{tails } \frac{43}{102}$$

Laplace smoothing:

- Easy to implement/remember

Back to our happy Laplace

Consider our previous 6-sided die.

- Roll the die $n = 12$ times.
- Observe: 3 ones, 2 twos, 0 threes, 3 fours, 1 fives, 3 sixes

Recall θ_{MLE} : $p_1 = 3/12, p_2 = 2/12, p_3 = 0/12, \quad \triangle!$
 $p_4 = 3/12, p_5 = 1/12, p_6 = 3/12$

What are your Laplace estimates for each roll outcome?

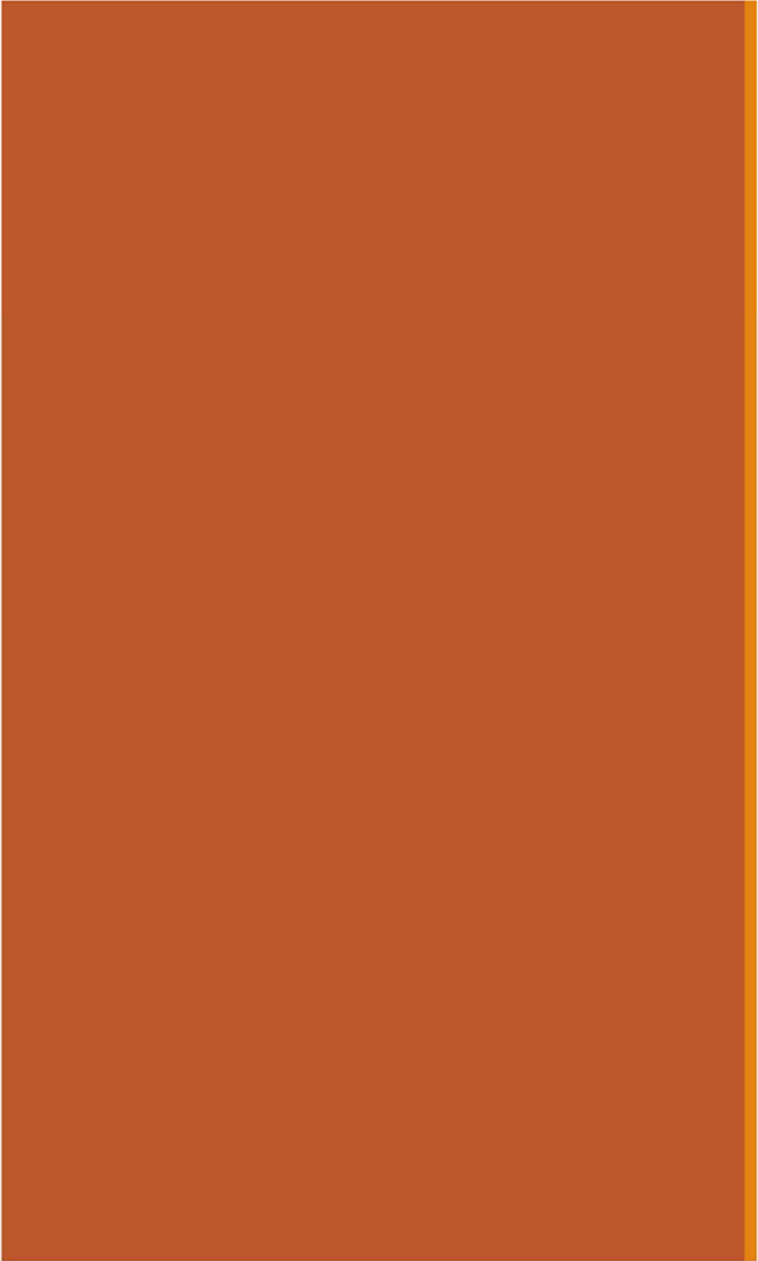
$$p_i = \frac{X_i + 1}{n + m}$$

$$X_3 = 0 \Rightarrow \frac{0+1}{12+6} = \frac{1}{18}$$

$$p_1 = 4/18, p_2 = 3/18, p_3 = 1/18, \quad \checkmark$$
$$p_4 = 4/18, p_5 = 2/18, p_6 = 4/18$$

Laplace smoothing:

- Easy to implement/remember
 - Avoids parameter estimation of 0
- often very important*



Other Conjugates

Conjugate distributions

MAP
estimator:

$$\theta_{MAP} = \arg \max_{\theta} f(\theta | X_1, X_2, \dots, X_n)$$

The **mode** of the
posterior distribution of θ

"ideal" model for prior and posterior

Distribution parameter	Conjugate distribution
Bernoulli p	Beta
Binomial p	Beta
Multinomial p_i	Dirichlet
Poisson λ	Gamma
Exponential λ	Gamma
Normal μ	Normal
Normal σ^2	Inverse Gamma

generalization of Beta

Multinomial is Multiple times the fun

Dirichlet($a_1, a_2, \dots, a_m$) is a conjugate for Multinomial.

- Generalizes Beta in the same way Multinomial generalizes Binomial:

$$f(x_1, x_2, \dots, x_m) = \frac{1}{B(a_1, a_2, \dots, a_m)} \prod_{i=1}^m x_i^{a_i-1}$$

Prior

Dirichlet($a_1, a_2, \dots, a_m$)

Saw $(\sum_{i=1}^m a_i) - m$ imaginary trials, with $a_i - 1$ of outcome i

Experiment

Observe $n_1 + n_2 + \dots + n_m$ new trials, with n_i of outcome i

Posterior

Dirichlet($a_1 + n_1, a_2 + n_2, \dots, a_m + n_m$)

MAP:

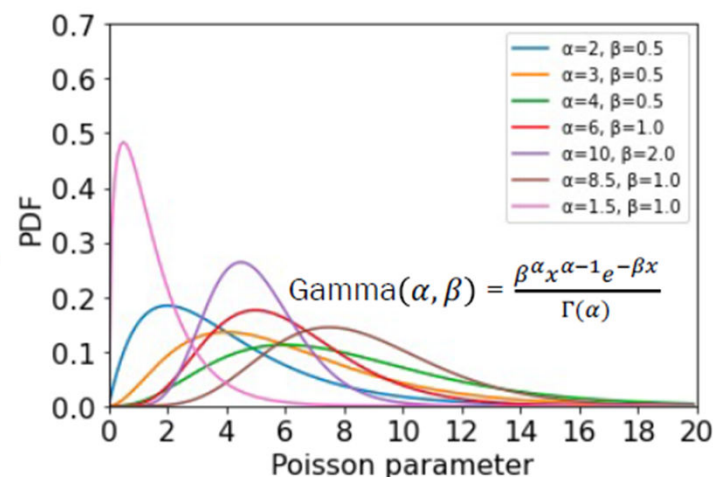
$$p_i = \frac{a_i + n_i - 1}{(\sum_{i=1}^m a_i) + (\sum_{i=1}^m n_i) - m}$$



Good times with Gamma

Gamma(α, β) is a conjugate for Poisson.

- Also conjugate for Exponential, but we won't delve into that
 - Mode of gamma: $(\alpha - 1)/\beta$
- Handwritten notes:*
 α : count events
 β : tracks the time that passed.



Prior

$$\theta \sim \text{Gamma}(\alpha, \beta) = \frac{\beta^\alpha x^{\alpha-1} e^{-\beta x}}{\Gamma(\alpha)}$$

Saw $\alpha - 1$ total imaginary events during β prior time periods

Experiment Observe n events during next k time periods

Posterior $(\theta | n \text{ events in } k \text{ periods}) \sim \text{Gamma}(\alpha + n, \beta + k)$

MAP:

Handwritten notes:
the new α
the new β

$$\theta_{MAP} = \frac{a + n - 1}{\beta + k}$$

MAP for Poisson

Gamma(α, β)
is conjugate for Poisson Mode: $\frac{\alpha-1}{\beta}$

Let λ be the average # of successes in a time period.

1. What does it mean to have a prior of $\theta \sim \text{Gamma}(11, 5)$?

Observe 10 imaginary events
in 5 time periods,
i.e., observe at Poisson rate = 2

Now perform the experiment and see 11 events in next 2 time periods.

2. Given your prior, what is the posterior distribution?
3. What is θ_{MAP} ?



MAP for Poisson

Gamma(α, β)
is conjugate for Poisson Mode: $\frac{\alpha-1}{\beta}$

Let λ be the average # of successes in a time period.

1. What does it mean to have a prior of $\theta \sim \text{Gamma}(11, 5)$?

Observe 10 imaginary events
in 5 time periods,
i.e., observe at Poisson rate = 2

Now perform the experiment and see 11 events in next 2 time periods.

2. Given your prior, what is the posterior distribution?

$(\theta | n \text{ events in } k \text{ periods}) \sim \text{Gamma}(22, 7)$

3. What is θ_{MAP} ?

$\theta_{MAP} = 3$, the updated Poisson rate